THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** $2\,Mg + O_2 \longrightarrow 2\,MgO$	Two Mg atoms combine with one O2 molecule; MgO is the formula of magnesium oxide.
2	**(c)** 10	Bases turn red litmus blue and have pH above 7.
3	**(d)** $Fe + CuSO_4$ solution	Fe is more reactive than Cu; others involve a less reactive metal.
4	**(a)** 4	O=C=O has two double bonds: 4 shared pairs.
5	**(a)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$	Ions are exchanged between the two compounds.
6	**(d)** 1.4	Stomach has hydrochloric acid.
7	**(c)** its easy loss of an electron	Sodium readily forms Na^+.
8	**(d)** long-chain carboxylic acids	Fatty acid salts.
9	**(a)** zinc is more reactive and displaces copper from its salt	Reactive metal displaces less reactive metal.
10	**(a)** magnesium hydroxide	$Mg(OH)_2$ neutralises acid.
11	**(b)** electrolysis of molten chlorides	They are too reactive for carbon reduction.
12	**(a)** catenation	Catenation.
13	**(a)** Displacement	More reactive Al displaces Fe from its oxide.
14	**(b)** HCl	Egg-shell is $CaCO_3$; acids release CO_2.
15	**(a)** applying grease	Paint/zinc cannot be used on a pan used for cooking; grease is suitable.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Mg is covered by a layer of magnesium oxide (and carbonate) (1); The layer is protective, so it must be rubbed off with sandpaper for burning (1).	1+1
17	Acids give H^+ ions only in the presence of water (1); Dry HCl has no H^+ ions free (1).	1+1
18	Mercury (1); Sodium (or potassium) (1).	1+1
19	Carbon has four valence electrons (1); Cannot easily lose or gain 4 electrons, so it shares them (1).	1+1
20	Paint forms a coating that keeps iron away from air and moisture (1); This prevents rusting (corrosion) (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Iron is more reactive than copper and displaces it (1); Blue $CuSO_4$ changes to pale green $FeSO_4$ (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1+1
22	Acid + metal -> salt + hydrogen (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen burns with a pop sound (1).	1+1+1
23	Hammer: metals are malleable, non-metals break into pieces (1); Circuit: metals complete the circuit and the bulb glows (1); Non-metals (except graphite) do not conduct (1).	1+1+1
24	n-butane $CH_3CH_2CH_2CH_3$ (1); isobutane (2-methylpropane) $CH_3CH(CH_3)CH_3$ (1); Both are C_4H_{10} (1).	1+1+1
	OR: Equation with equal number of atoms of each element on both sides (1); Balancing by hit and trial (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1).	1+1+1
25	Electrolysis of brine (aqueous NaCl) (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Products: NaOH, chlorine, hydrogen (1).	1+1+1
	OR: Calcination: $ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Reduction: $ZnO + C \longrightarrow Zn + CO$ (1); Carbon acts as reducing agent (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Equation with equal atoms of each element on both sides (1); Fe3O4 contains 3 Fe, so put 3 before Fe (1); 4 O atoms in Fe3O4 need 4 H2O (1); Put 4 before H2 to balance hydrogen (1); $3\,Fe\,(s) + 4\,H_2O\,(g) \longrightarrow Fe_3O_4\,(s) + 4\,H_2\,(g)$ (1).	1+1+1+1+1
	OR: Bubbles on the surface of zinc (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Gas burns with a pop sound near a flame (1); Test tube becomes warm (exothermic) (1); Acid + metal -> salt + hydrogen (1).	1+1+1+1+1
27	$4\,Na + O_2 \longrightarrow 2\,Na_2O$ (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (amphoteric) (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (black) (1); Amphoteric oxides react with both acid and base (1).	1+1+1+1+1
	OR: Carbon has 4 valence electrons (1); Forms covalent bonds by sharing (1); Catenation forms chains and rings (1); Tetravalency: bonds with 4 atoms (1); CH_4: C shares 4 pairs with 4 H (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 16	3n + 1 = 16.
2	**(b)** $BaSO_4$	Barium sulphate is insoluble in water.
3	**(c)** 5.5	Enamel dissolves below pH 5.5.
4	**(c)** calcium	Ca reacts less violently and floats.
5	**(a)** formation of micelles	Hydrophobic tails trap oil in micelles.
6	**(c)** 2	$H_2 + Cl_2 \longrightarrow 2\,HCl$
7	**(a)** basic solution	Pink in bases, colourless in acids.
8	**(a)** impure copper	Impure copper dissolves.
9	**(d)** four valence electrons	Difficult to gain or lose 4 electrons.
10	**(d)** Calcium oxide (quick lime)	Quick lime reacts with water to give slaked lime used for whitewashing.
11	**(a)** 16 mL	Proportion: 8 x 20/10 = 16 mL.
12	**(b)** calcium	A metal forms an ionic oxide; calcium oxide is ionic and dissolves.
13	**(b)** C_2H_4	Ethene has a double bond.
14	**(b)** gain of oxygen or loss of hydrogen	Definition of oxidation (in terms of oxygen/hydrogen).
15	**(c)** tartaric acid	Tamarind contains tartaric acid.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Acid forms H^+ (hydronium) ions in water (1); Moving ions carry current (1).	1+1
17	Mixture of conc. HCl and conc. HNO_3 in 3 : 1 (1); It dissolves gold and platinum (1).	1+1
18	Atom or group that decides chemical properties (1); -OH (alcohol) (1).	1+1
19	Blue colour fades and reddish-brown copper is deposited (1); Displacement reaction (1).	1+1
20	Calcium oxychloride (1); $CaOCl_2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ethanol is oxidised by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ (1); $CH_3CH_2OH \longrightarrow CH_3COOH$ (1); Oxidising agent adds oxygen (1).	1+1+1
22	X is calcium oxide (quick lime) (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Exothermic reaction giving slaked lime (1).	1+1+1
23	$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$ (1); Free H^+ cannot exist alone; it combines with water (1); H_3O^+ is the hydronium ion (1).	1+1+1
24	Metals conduct heat and electricity; non-metals do not (graphite exception) (1); Metal oxides are basic (1); Non-metal oxides are acidic (1).	1+1+1
	OR: Saturated: single bonds only; C_2H_6 (1); Unsaturated: double or triple bonds; C_2H_4 (1); Unsaturated show addition reactions (1).	1+1+1
25	$Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ (1); Exchange of ions occurs between the compounds (1); Yellow precipitate of PbI_2 forms (1).	1+1+1
	OR: Salt, water and carbon dioxide are formed (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); CO_2 turns lime water milky (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Oxidation: gain of oxygen/loss of hydrogen; reduction: loss of oxygen/gain of hydrogen (1); C is oxidised (1); ZnO is reduced (1); ZnO is the oxidising agent (1); C is the reducing agent (1).	1+1+1+1+1

	OR: Electrolysis of brine (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Cl2 at anode, H2 at cathode (1); Uses: NaOH in soap; Cl_2 in water treatment (1); H_2 as fuel/ margarine (1).	1+1+1+1+1
27	Slow attack of air and moisture on metals (1); Rusting needs both air and water (1); Painting and oiling (1); Galvanisation and chrome plating (1); Alloying and anodising (1).	1+1+1+1+1
	OR: $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Conc. H_2SO_4 catalyst (1); Esters have fruity smell (1); Used in perfumes and flavours (1); $CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** copper	Copper is below hydrogen in the series.
2	**(b)** calcium and magnesium salts forming scum	Insoluble Ca/Mg salts of fatty acids.
3	**(c)** 2 : 1	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
4	**(c)** red	Red in acid, yellow in base.
5	**(d)** acidified copper sulphate solution	Cu^{2+} ions move to the cathode.
6	**(c)** 2	n-butane and isobutane.
7	**(b)** remove the protective oxide layer from its surface	The oxide coating prevents the metal from burning.
8	**(a)** antacid	Antacids neutralise excess stomach acid.
9	**(d)** zinc is more reactive than tin	Zinc would react with acids in food.
10	**(a)** C_4H_8	Homologues differ by CH_2.
11	**(b)** H_2	H_2 gains oxygen to form H_2O.
12	**(a)** methanoic acid	Methanoic acid causes the burning pain.
13	**(b)** $MgCl_2$	Metal + non-metal compound is ionic.
14	**(a)** an alcohol	Esterification.
15	**(a)** thermal decomposition	Heat breaks down limestone.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$Na \longrightarrow Na^+ + e^-$ and $Cl + e^- \longrightarrow Cl^-$ (1); $Na^+ + Cl^- \longrightarrow NaCl$ (1).	1+1
17	Acid reacts with alcohol to give ester and water (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1).	1+1
18	Skeleton: $H_2 + Cl_2 \longrightarrow HCl$ (1); Balanced: $H_2 + Cl_2 \longrightarrow 2\,HCl$ (1).	1+1
19	Dilution is highly exothermic (1); Adding water to acid may splash hot acid and cause burns (1).	1+1
20	It reacts very vigorously with oxygen and moisture (1); Kerosene keeps it away from air/water preventing fire (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1); $Ca + 2\,H_2O \longrightarrow Ca(OH)_2 + H_2$ (1); Calcium floats due to hydrogen bubbles (1).	1+1+1
22	C_2H_4, $CH_2{=}CH_2$ (1); Contains a carbon-carbon double bond (1); Can add hydrogen or halogen (1).	1+1+1
23	Combination: two or more substances form one product, e.g. $CaO + H_2O \longrightarrow Ca(OH)_2$ (2); Decomposition: one compound breaks into simpler substances, e.g. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1).	2+1
24	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Used for bleaching cotton/linen (1); Used to disinfect drinking water (1).	1+1+1
	OR: Mercury is a very unreactive metal (1); $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$ (1); $2\,HgO \longrightarrow 2\,Hg + O_2$ (1).	1+1+1
25	$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (2).	1+2
	OR: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Hydrogen (cathode) is twice oxygen by volume (1); Because water has two hydrogen atoms for one oxygen atom (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Combination reaction (1); Product is a white powder, magnesium oxide (1); Its solution turns red litmus blue (basic) (1).	1+1+1+1+1
	OR: Set up circuit with bulb and two electrodes (1); HCl solution: bulb glows (1); Glucose solution: bulb does not glow (1); Acids give H^+ ions in water (1); Glucose does not give ions (1).	1+1+1+1+1

27	Order: Na > Mg > Al > Zn > Fe > Pb > Cu > Ag > Au (1); Na/K react violently with cold water (1); Mg reacts with hot water, Al/Zn/Fe with steam (1); Cu, Ag, Au do not react (1); Metals above hydrogen displace it from acids (1).	1+1+1+1+1
	OR: Same molecular formula C_5H_{12} (1); n-pentane (1); 2-methylbutane (isopentane) (1); 2,2-dimethylpropane (neopentane) (1); Three isomers (1).	1+1+1+1+1

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